



# Modelling Financial Markets

## Chapter 5: Conditional Volatility — GARCH Models



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## Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FIGARCH: Fractionally Integrated GARCH
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-M: GARCH-in-Mean
- ▣ GARCH-MIDAS: GARCH with Mixed Data Sampling (long-run component)
- ▣ GARCH-N: GARCH with Normal innovations
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ GED: Generalised Error Distribution
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)

## Acronyms used in this material (2/2)

- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ IGARCH: Integrated GARCH
- ▣ LM: Lagrange Multiplier
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ QQ: Quantile–Quantile (plot)
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index
- ▣ XAU: gold, ISO 4217 code (one troy ounce)

## Outline

Motivation

From ARCH to GARCH

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## Volatility Comes in Bursts



Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- Crashes are not isolated days: large moves are followed by large moves
  - ▶ “Large changes tend to be followed by large changes, of either sign” [Mandelbrot \(1963\)](#)
- This **volatility clustering** is stylised fact 6 of Chapter 1
- Volatility enters every risk number: Value at Risk (Chapter 7), portfolio weights (Chapter 4), option prices
- Question: can we turn the clustering into a **forecast** of tomorrow’s risk, and do valid inference on it?

## Data Used in This Chapter

| Series                 | Owner / source               | Start | Obs.  | Obs. per year |
|------------------------|------------------------------|-------|-------|---------------|
| S&P 500                | S&P Dow Jones Indices        | 2000  | 6 714 | 251.4         |
| BET                    | Bucharest Stock Exchange     | 2000  | 6 670 | 249.8         |
| BET-TR                 | Bucharest Stock Exchange     | 2014  | 2 992 | 249.7         |
| Bitcoin (BTC/USD)      | crypto exchanges (composite) | 2014  | 4 384 | 365.3         |
| EUR/RON reference rate | BNR                          | 2005  | 5 352 | 252.4         |
| Gold (XAU/USD)         | spot market                  | 2000  | 6 922 | 259.2         |

- Daily closes up to 18 September 2026; returns  $r_t = 100 (\ln P_t - \ln P_{t-1})$ , in %
- Equity indices and gold: weekdays only; gold is annualised with its actual frequency
- Bitcoin trades 7 days a week; the VIX index of the Chicago Board Options Exchange is used as a benchmark

## Prerequisites Recap: Clustering and Heavy Tails (Chapter 1)

- Heavy tails: excess kurtosis S&P 500 10.7, BET 11.1, Bitcoin 12.0 (0 for the Normal distribution)
  - ▶ Annual volatility: Bitcoin 66.6%, S&P 500 19.2%, BET 22.1%
- S&P 500: lag-1 autocorrelation of  $r_t$  is -0.099, of  $r_t^2$  is 0.313
  - ▶ All 50 autocorrelations of  $r_t^2$  up to lag 50 lie outside  $\pm 0.024$ ; still 0.053 at lag 50
- The sign of tomorrow's return is almost unpredictable; its size is not: the object to model is  $\text{Var}(r_t \mid \mathcal{F}_{t-1})$
- Charts of both facts: Quantlet below and Chapter 1

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## Testing for ARCH Effects

- ▣ **McLeod–Li** [McLeod & Li \(1983\)](#): Ljung–Box  $Q(10)$  on  $r_t^2$ ,  $\chi_{10}^2$  under no clustering
- ▣ **Engle's LM test** [Engle \(1982\)](#):  $nR^2$  from regressing  $\hat{\varepsilon}_t^2$  on  $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-5}^2$ ,  $\chi_5^2$ ;  $\hat{\varepsilon}_t = r_t - \bar{r}$
- ▣ The alternative is one-sided ( $\alpha_i \geq 0$ ): the two-sided  $\chi^2$  test wastes power
  - ▶ [Lee & King \(1993\)](#): one-sided score test built on the sum of the score components,  $N(0, 1)$  under  $H_0$ , rejects in the upper tail only
- ▣ Against GARCH(1,1) the null  $\alpha = 0$  leaves  $\beta$  unidentified (Davies problem)
  - ▶ The ARCH( $q$ ) LM statistic is also the score test against GARCH( $p, q$ ) [Lee & King \(1993\)](#): the same statistic serves both alternatives

## ARCH Effects in Six Markets

| Series  | Vol. (% p.a.) | Excess kurt. | $\hat{\rho}_1(r^2)$ | $Q(10)$ on $r^2$ | LM(5) |
|---------|---------------|--------------|---------------------|------------------|-------|
| S&P 500 | 19.2          | 10.7         | 0.313               | 6133             | 1679  |
| BET     | 22.1          | 11.1         | 0.330               | 2440             | 983   |
| BET-TR  | 15.4          | 17.8         | 0.171               | 714              | 372   |
| Bitcoin | 66.6          | 12.0         | 0.137               | 213              | 115   |
| EUR/RON | 4.4           | 15.1         | 0.342               | 2420             | 872   |
| Gold    | 17.6          | 7.7          | 0.110               | 744              | 330   |

- 5% critical values:  $\chi_{10}^2$ : 18.3;  $\chi_5^2$ : 11.1; every statistic is far above them (all p-values  $< 0.001$ )
- Clustering is universal: equities, crypto, a managed exchange rate and gold
- Weakest (but still decisive) for Bitcoin and gold, strongest for the S&P 500

## The ARCH Model of Engle (1982)

- Setting:  $r_t = \mu + \varepsilon_t$ ,  $\varepsilon_t = \sigma_t z_t$ , innovations  $z_t$  i.i.d. with mean 0 and variance 1
  - ▶ Conditional variance  $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1})$  ( $\mathcal{F}_{t-1}$ : information at  $t-1$ ); unconditional  $\sigma^2 = \mathbb{E}\sigma_t^2$
- **ARCH( $q$ )**:  $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \dots + \alpha_q \varepsilon_{t-q}^2$  [Engle \(1982\)](#)
  - ▶  $\omega > 0$ ,  $\alpha_i \geq 0$ : the variance is always positive
- A large shock yesterday ( $\varepsilon_{t-1}^2$  large) raises today's variance: clustering by construction
- ARCH(1): unconditional variance  $\sigma^2 = \omega/(1 - \alpha_1)$ , finite if  $\alpha_1 < 1$ 
  - ▶ Kurtosis  $3(1 - \alpha_1^2)/(1 - 3\alpha_1^2) > 3$  for Normal  $z_t$ : heavy tails appear even with Normal innovations

## Robert Engle and the 2003 Nobel Prize



Robert F. Engle (2022)

Photo: Jon Demske (2022); CC

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Clive W.J. Granger (2008)

Photo: Olaf Storbeck (2008); CC

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- ▣ Nobel Prize 2003: Engle “for methods of analyzing economic time series with time-varying volatility (ARCH)”
  - Shared with Clive Granger (cointegration)
- ▣ Nobel lecture: Engle (2004), on risk and volatility in financial practice
- ▣ Tim Bollerslev, Engle’s doctoral student, generalised ARCH to GARCH in 1986

## From ARCH to GARCH (Bollerslev, 1986)

- Problem with ARCH: clustering in real data lasts for months, so  $q$  must be large (many parameters)
- **GARCH(1,1)** [Bollerslev \(1986\)](#):  $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ 
  - ▶  $\omega > 0, \alpha \geq 0, \beta \geq 0$ ;  $\alpha$ : reaction to news;  $\beta$ : memory of past variance
- Substituting backwards:  $\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j \geq 1} \beta^{j-1} \varepsilon_{t-j}^2$ 
  - ▶ An ARCH( $\infty$ ) with geometrically declining weights, using only three parameters
- GARCH( $p, q$ ) adds more lags; in practice GARCH(1,1) is the workhorse [Hansen & Lunde \(2005\)](#)

## GARCH(1,1) as an ARMA Model for Squared Shocks

- Define  $\nu_t = \varepsilon_t^2 - \sigma_t^2$ , the surprise in the squared shock ( $\mathbb{E}[\nu_t | \mathcal{F}_{t-1}] = 0$ )
- Then  $\varepsilon_t^2 = \omega + (\alpha + \beta) \varepsilon_{t-1}^2 + \nu_t - \beta \nu_{t-1}$ 
  - ▶ An ARMA(1,1) in  $\varepsilon_t^2$  with autoregressive coefficient  $\alpha + \beta$
- Consequence: the autocorrelations of  $\varepsilon_t^2$  decay geometrically,  $\rho_k = \rho_1(\alpha + \beta)^{k-1}$ 
  - ▶  $\rho_1 = \alpha(1 - \beta^2 - \alpha\beta)/(1 - \beta^2 - 2\alpha\beta)$  if  $\mathbb{E}\varepsilon_t^4 < \infty$  (Seminar 5, A6)
  - ▶ S&P 500, GARCH(1,1) with Normal innovations:  $\rho_1 = 0.356$ ,  $\rho_{10} = 0.298$ ; sample 0.313 and 0.248
- $\alpha + \beta$  is therefore called the **persistence** of volatility

## Stationarity and the Unconditional Variance

- **Covariance stationarity:**  $\alpha + \beta < 1$ ; then
  - ▶  $\sigma^2 = \mathbb{E}[\sigma_t^2] = \frac{\omega}{1 - \alpha - \beta}$
- Derivation: take expectations in the recursion,  $\mathbb{E}\sigma_t^2 = \omega + (\alpha + \beta)\mathbb{E}\sigma_{t-1}^2$ , and set  $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2$
- **IGARCH** (integrated GARCH):  $\alpha + \beta = 1$ ; the unconditional variance is infinite
  - ▶ Yet the process is still **strictly** stationary: the condition is  $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$  [Nelson \(1990\)](#)
- Variance targeting: fix  $\omega = \hat{\sigma}^2(1 - \alpha - \beta)$  with the sample variance  $\hat{\sigma}^2$

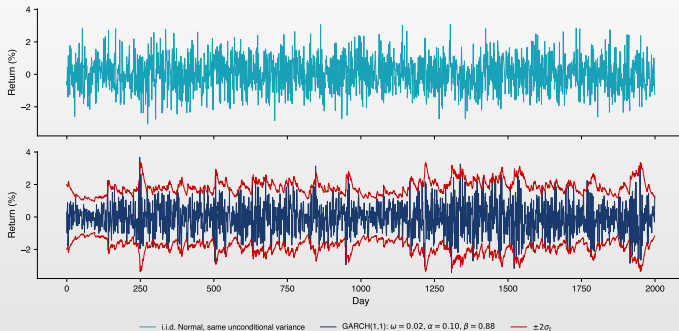
## Persistence and Half-Life

- Multi-step forecast:  $\mathbb{E}[\sigma_{t+h}^2 \mid \mathcal{F}_t] = \sigma^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \sigma^2)$ 
  - ▶ The gap to the long-run variance shrinks by the factor  $\alpha + \beta$  each day
- **Half-life:** number of days after which half of a variance shock has gone
  - ▶  $(\alpha + \beta)^h = 0.5 \Rightarrow h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$
- Examples:  $\alpha + \beta = 0.95$ : 13.5 days; 0.99: 69 days; 0.995: 138 days
- Small changes in  $\alpha + \beta$  near 1 imply very different half-lives: estimation uncertainty matters (Seminar 5)

## Fourth Moment and Kurtosis

- With Normal  $z_t$ , the kurtosis of  $\varepsilon_t$  is  $K = \frac{3[1 - (\alpha + \beta)^2]}{1 - (\alpha + \beta)^2 - 2\alpha^2}$ 
  - ▶ Finite only if  $(\alpha + \beta)^2 + 2\alpha^2 < 1$  (Seminar 1, Exercise A9)
- S&P 500, GARCH(1,1) with Normal innovations:  $\hat{\alpha} = 0.1206$ ,  $\hat{\beta} = 0.8599$ 
  - ▶  $(\hat{\alpha} + \hat{\beta})^2 + 2\hat{\alpha}^2 = 0.9904$ : just below 1, so  $K = 12.1$ ; sample kurtosis of the returns: 13.7
- $K$  is ill-conditioned near the moment boundary: the denominator is  $1 - 0.9904$ 
  - ▶ Rounding to  $\hat{\alpha} = 0.121$ ,  $\hat{\beta} = 0.860$  gives 0.9916 and  $K = 13.5$ : report such functions from unrounded estimates
- Clustering alone generates much of the fat tails; the rest comes from heavy-tailed  $z_t$  (Section 6)

## Simulation: GARCH Produces Clustering



- Top: i.i.d. Normal returns; bottom: GARCH(1,1) with the same unconditional variance (1) and  $\pm 2\sigma_t$  bands

## Interpretation: Clustering from a Three-Parameter Rule

- Same unconditional variance, very different paths: quiet stretches and bursts appear only in the GARCH path
- On a long simulated path (one million days)
  - ▶ Excess kurtosis 2.47 (theory 3.06); lag-1 autocorrelation of  $\varepsilon_t^2$ : 0.26
  - ▶ The sample kurtosis converges slowly: its precision depends on the eighth moment, infinite for these parameters
- Normal innovations, yet fat-tailed returns: the mixture of variances creates the tails

## EWMA: a GARCH Special Case

- **EWMA** J.P. Morgan/Reuters (1996):  $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$ , with  $\lambda = 0.94$  for daily data
  - ▶ Weights  $(1 - \lambda)\lambda^{j-1}$  on past squared returns: the most recent days count most
- EWMA is an IGARCH(1,1) with  $\omega = 0$ ,  $\alpha = 1 - \lambda$ ,  $\beta = \lambda$ : no parameter to estimate
- Price of simplicity: no mean reversion
  - ▶ The forecast for any horizon equals today's variance; GARCH pulls it towards  $\sigma^2$
- We use EWMA as a benchmark in the forecast comparison of Section 7

## Recap: ARCH and GARCH

### Key points

- ▣  $\sigma_t^2$  is the variance given yesterday's information; it moves with the size of recent shocks
- ▣  $\text{GARCH}(1,1) = \text{ARCH}(\infty)$  with three parameters;  $\alpha + \beta$  is the persistence
- ▣  $\alpha + \beta < 1$ : finite long-run variance  $\omega/(1 - \alpha - \beta)$  and half-life  $\ln 0.5 / \ln(\alpha + \beta)$
- ▣ Clustering creates fat tails, even with Normal innovations
- ▣ EWMA is the IGARCH special case with  $\omega = 0$

## Maximum Likelihood

- ▣ Parameters  $\theta = (\mu, \omega, \alpha, \beta)$ ; given  $\theta$ , the recursion produces  $\sigma_1^2(\theta), \dots, \sigma_n^2(\theta)$
- ▣ Normal log-likelihood:  $\ell(\theta) = -\frac{1}{2} \sum_{t=1}^n [\ln 2\pi + \ln \sigma_t^2 + \varepsilon_t^2 / \sigma_t^2]$ 
  - ▶ Each day contributes the density of  $\varepsilon_t$  given the past: the likelihood factorises in time
- ▣ No closed form: numerical optimisation under  $\omega > 0, \alpha, \beta \geq 0, \alpha + \beta < 1$
- ▣ Start-up:  $\sigma_1^2$  is set to the sample variance (its effect fades at rate  $\beta$ )
- ▣ Scale: returns in %, so that the variance is of order 1
  - ▶ EUR/RON (daily s.d. about 0.28%) is fitted on 10  $r_t$ : unscaled, the optimiser stopped at a local optimum

## QMLE: Consistency and Asymptotic Normality

- **Gaussian QMLE:** maximise the Normal likelihood even if  $z_t$  is not Normal [Bollerslev & Wooldridge \(1992\)](#)
- **Consistency** (GARCH(1,1)) [Berkas, Horváth & Kokoszka \(2003\)](#), [Francq & Zakoïan \(2004\)](#)
  - ▶ Strict stationarity  $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$ ;  $\theta_0$  in a compact set;  $z_t^2$  non-degenerate
  - ▶ No moment condition on  $\varepsilon_t$ : IGARCH and infinite-variance cases are covered
- **Asymptotic normality:** add  $\theta_0$  interior and  $\kappa_z = \mathbb{E} z_t^4 < \infty$ 
  - ▶  $\sqrt{n}(\hat{\theta} - \theta_0) \rightarrow N(0, (\kappa_z - 1)J^{-1})$ ,  $J = \mathbb{E}[g_t g_t']$ ,  $g_t = \partial \ln \sigma_t^2 / \partial \theta$
  - ▶ Holds also for  $\alpha + \beta \geq 1$ : [Lumsdaine \(1996\)](#) (IGARCH), [Jensen & Rahbek \(2004\)](#) (nonstationary)
- Sandwich  $\hat{A}^{-1} \hat{B} \hat{A}^{-1}$  estimates  $(\kappa_z - 1)J^{-1}$ ; the inverse Hessian  $\hat{A}^{-1}$  estimates  $2J^{-1}$ 
  - ▶  $A$ : minus the expected Hessian;  $B$ : variance of the score; robust / classical s.e. =  $\sqrt{(\kappa_z - 1)/2}$ , 1 under Normality

## QMLE in Our Data: When the Theory Holds and When It Fails

- S&P 500, GARCH(1,1) with Normal likelihood: residual kurtosis  $\hat{\kappa}_z = 4.73$ 
  - ▶ Theory: robust / classical s.e. =  $\sqrt{(4.73 - 1)/2} = 1.37$ ; observed: 1.37 for  $\hat{\alpha}$ , 1.37 for  $\hat{\beta}$
- Bitcoin: GARCH-t gives  $\hat{\nu} = 3.18 < 4$ , so  $\mathbb{E}z_t^4 = \infty$  and the normality condition fails
  - ▶ Residual kurtosis does not settle: 19.8 on the first half, 14.6 on the full sample; s.e. ratios 2.51 and 2.01
- Hall & Yao (2003): then the Gaussian QMLE converges slower than  $\sqrt{n}$ , to a non-Gaussian limit: robust  $t$ -statistics are not valid
- Remedies: a heavy-tailed quasi-likelihood Fan, Qi & Xiu (2014), or the Student-t MLE with inference by parametric bootstrap

## Heavy-Tailed Innovations

- **Student-t** [Bollerslev \(1987\)](#):  $z_t = \sqrt{(\nu - 2)/\nu} t_\nu$ , unit variance for  $\nu > 2$ 
  - ▶  $\nu$  (degrees of freedom): the smaller  $\nu$ , the heavier the tails; the fourth moment exists only for  $\nu > 4$
- **Skewed-t** [Hansen \(1994\)](#): adds an asymmetry parameter  $\lambda \in (-1, 1)$ 
  - ▶  $\lambda < 0$ : the left tail is longer (large losses more likely than large gains)
- **GED** [Nelson \(1991\)](#): density  $\propto \exp(-\frac{1}{2}|z/\lambda_\nu|^\nu)$ 
  - ▶  $\nu = 2$ : the Normal distribution;  $\nu = 1$ : Laplace;  $\nu < 2$ : heavier tails ( $\lambda_\nu$  gives unit variance)

## Testing on the Boundary of the Parameter Space

- $LR = 2(\ell_1 - \ell_0) \rightarrow \chi_m^2$  only for interior restrictions (e.g. GJR vs GARCH:  $\gamma = 0$ ,  $m = 1$ )
  - ▶ Non-nested models: **AIC** =  $-2\ell + 2k$ , **BIC** =  $-2\ell + k \ln n$  ( $k$  parameters); lower is better
- GARCH restrictions  $\alpha, \beta \geq 0$ ,  $1/\nu \geq 0$ ,  $\alpha + \beta \leq 1$  put many nulls **on the boundary**
  - ▶ One parameter on the boundary:  $LR \rightarrow \frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$  [Self & Liang \(1987\)](#), [Andrews \(2001\)](#); 5% critical value 2.71, not 3.84
  - ▶ Cases: Normal vs Student-t ( $1/\nu = 0$ ); GJR with  $\alpha = 0$ ; IGARCH; no ARCH ( $\alpha = 0$ ,  $\beta$  unidentified)
- Wald  $t$ -tests of a parameter estimated at the bound (e.g.  $\hat{\alpha} = 0$ ) are not  $N(0, 1)$ 
  - ▶ The parametric bootstrap from  $\hat{\theta}$  is inconsistent when  $\theta_0$  is on the boundary [Andrews \(2000\)](#): simulate under  $H_0$  instead

## Boundary Tests on Our Data

- Normal vs Student-t innovations, GARCH(1,1):  $LR = 311.0$  (S&P 500), 1082.0 (Bitcoin)
  - ▶ Far above 2.71: heavy tails are decisive; the mixture matters only in borderline cases
- GJR-t, S&P 500:  $\hat{\alpha} = 0.000$ ; its  $t$ -test is not standard, and the  $\chi_1^2$  law of the LR for  $\gamma = 0$  assumes  $\alpha_0 > 0$
- IGARCH,  $H_0: \alpha + \beta = 1$  (GARCH-t):  $LR = 1.11$  (S&P 500), 1.14 (BET)
  - ▶ S&P 500:  $p = 0.29$  ( $\chi_1^2$ ), 0.15 (mixture), 0.17 (parametric bootstrap under  $H_0$ ,  $B = 999$ )
  - ▶ BET:  $p = 0.28$ , 0.14 and 0.17
- Under  $H_0$ , 52% of the re-estimates sit at  $\alpha + \beta = 1$ , as the mixture predicts: a unit root in variance cannot be rejected for either index

## S&amp;P 500: GARCH(1,1)-t Estimates

| Parameter        | Estimate | Robust s.e. | Classical s.e. | Ratio |
|------------------|----------|-------------|----------------|-------|
| $\omega$         | 0.0157   |             |                |       |
| $\alpha$         | 0.123    | 0.011       | 0.010          | 1.06  |
| $\beta$          | 0.872    | 0.010       | 0.010          | 1.08  |
| $\nu$            | 6.26     | 0.48        |                |       |
| $\alpha + \beta$ | 0.9947   |             |                |       |

- Daily returns in %, 2000–2026; mean  $\hat{\mu} = 0.079\%$  per day
- Half-life 131 days; long-run volatility  $\sqrt{252 \hat{\omega} / (1 - \hat{\alpha} - \hat{\beta})} = 27.3\%$  p.a.

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## Interpretation: S&P 500 Estimates

- ▣  $\hat{\beta} = 0.872$ : most of today's variance is yesterday's variance;  $\hat{\alpha} = 0.123$ : news adds a moderate update
- ▣ Persistence 0.9947: shocks to volatility last months, not days
- ▣ The long-run volatility (27.3%) exceeds the sample volatility (19.2%)
  - ▶ Near  $\alpha + \beta = 1$ ,  $\omega/(1 - \alpha - \beta)$  divides by a tiny number and is imprecise
- ▣  $\hat{\nu} = 6.26$ : innovations clearly heavier-tailed than the Normal distribution
- ▣ Robust standard errors are about 5–7% larger than the classical ones for  $\alpha$  and  $\beta$

## S&P 500: Twelve Models Compared by BIC

| $\Delta\text{BIC}$ | $N(0, 1)$ | Student-t | Skewed-t | GED   |
|--------------------|-----------|-----------|----------|-------|
| GARCH(1,1)         | 648.9     | 346.7     | 307.4    | 339.8 |
| GJR-GARCH(1,1)     | 394.7     | 119.9     | 49.6     | 141.6 |
| EGARCH(1,1)        | 372.9     | 74.8      | 0.0      | 111.4 |

- $\Delta\text{BIC}$ : BIC minus the lowest BIC among the 12 models (0 = preferred model)
- Two choices matter: an asymmetric variance equation and a heavy-tailed, skewed innovation
- Moving from Normal to Student-t innovations lowers BIC by about 300; adding asymmetry lowers it by a further 230–270
- Winner: EGARCH with skewed-t innovations; Sections 5–6 explain why

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## Recap: Estimation

### Key points

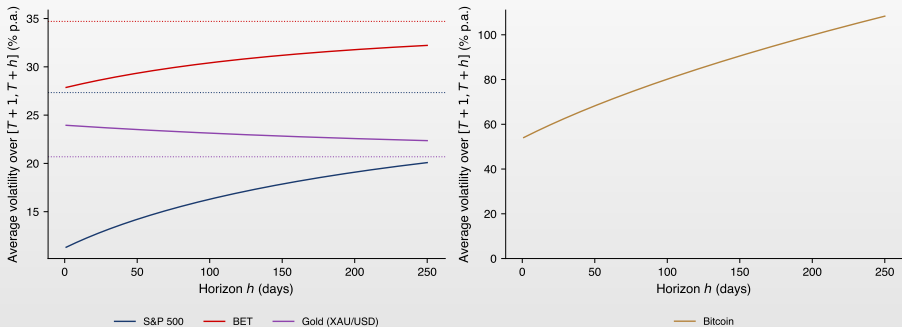
- ▣ GARCH is estimated by maximum likelihood; the likelihood is built day by day from the recursion
- ▣ QMLE is consistent without moment conditions; its  $\sqrt{n}$  normality needs  $\mathbb{E}z_t^4 < \infty$ , which fails for Bitcoin
- ▣ Student-t, skewed-t and GED capture the tails that GARCH alone leaves
- ▣ Nulls on the boundary ( $1/\nu = 0$ ,  $\alpha = 0$ ): LR against  $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ ; IGARCH: simulate under  $H_0$
- ▣ Rescale low-volatility series; compare non-nested models with AIC/BIC

## Worked Example: One-Day and Ten-Day Forecasts

- S&P 500, GARCH(1,1)-t:  $\hat{\omega} = 0.0157$ ,  $\hat{\alpha} = 0.123$ ,  $\hat{\beta} = 0.872$ ,  $\hat{\mu} = 0.079$
- Last day  $T = 18.09.2026$ :  $\hat{\varepsilon}_T = r_T - \hat{\mu} = 0.088$ , filtered variance  $\hat{\sigma}_T^2 = 0.565$ 
  - ▶  $\hat{\sigma}_{T+1}^2 = \hat{\omega} + \hat{\alpha}\hat{\varepsilon}_T^2 + \hat{\beta}\hat{\sigma}_T^2 = \hat{\omega} + 0.0009 + 0.493 = 0.509$ , i.e. 11.3% p.a.
- Long-run variance  $\hat{\sigma}^2 = \hat{\omega}/(1 - \hat{\alpha} - \hat{\beta}) = 2.97$ ; day 10:  $\hat{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^9(\hat{\sigma}_{T+1}^2 - \hat{\sigma}^2)$ 
  - ▶  $= 2.97 + 0.953 \times (0.509 - 2.97) = 0.624$
- 10-day variance  $\sum_{h=1}^{10} \hat{\sigma}_{T+h}^2 = 5.67$ , volatility 2.38%
  - ▶ The square-root-of-time rule gives  $\sqrt{10} \times 0.714 = 2.26\%$ : too low, because  $\hat{\sigma}_{T+1}^2 < \hat{\sigma}^2$  and volatility is forecast to rise

 MFM\_ch5\_garch\_estimation

## Volatility Term Structure



Forecast from 18.09.2026 of the average volatility over the next  $h$  days (GARCH(1,1)-t); dotted: long-run level

## Interpretation: Mean Reversion or Not

- ▣ S&P 500: from 11.3% (next day) to 20.1% (one year), slowly rising towards the long-run level
- ▣ Gold: today above its long run, so the curve declines, from 23.9% to 22.4%
- ▣ Bitcoin and EUR/RON:  $\hat{\alpha} + \hat{\beta} = 1$  (IGARCH), no long-run level
  - ▶ The variance forecast grows by  $\hat{\omega}$  per day: Bitcoin from 54.1% to 108.3% over one year
  - ▶ An artefact of the unit-root estimate, not a prediction to trade on
- ▣ Long-horizon GARCH forecasts inherit all the uncertainty about  $\alpha + \beta$

## GARCH(1,1)-t in Six Markets

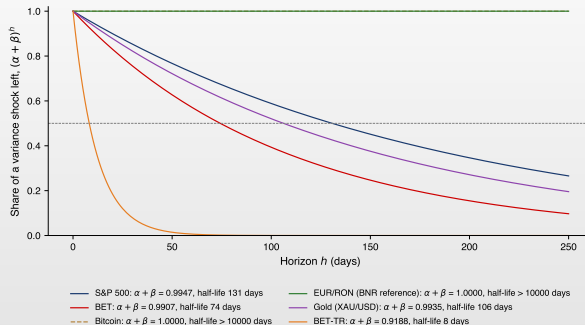
| Series  | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\alpha} + \hat{\beta}$ | $h_{1/2}$ (days) | $\hat{\nu}$ | Long-run vol. |
|---------|----------------|----------------|---------------|------------------------------|------------------|-------------|---------------|
| S&P 500 | 0.0157         | 0.123          | 0.872         | 0.9947                       | 131              | 6.26        | 27.3          |
| BET     | 0.0448         | 0.192          | 0.799         | 0.9907                       | 74               | 5.01        | 34.7          |
| BET-TR  | 0.0755         | 0.206          | 0.713         | 0.9188                       | 8                | 5.02        | 15.2          |
| Bitcoin | 0.1937         | 0.109          | 0.891         | 1.0000                       | $\infty$         | 3.18        | –             |
| EUR/RON | 0.0000         | 0.123          | 0.877         | 1.0000                       | $\infty$         | 4.20        | –             |
| Gold    | 0.0107         | 0.049          | 0.944         | 0.9935                       | 106              | 4.58        | 20.7          |

- Long-run volatility in % p.a. (annualised with each series' own frequency); – when  $\hat{\alpha} + \hat{\beta} \geq 0.999$
- EUR/RON:  $\hat{\omega}$  is 0.0000 at four decimals (the rate moves by fractions of a ban on most days)

## Interpretation: Six Volatility Personalities

- Equities (S&P 500, BET) and gold: persistence 0.99–0.995, half-lives of 3–6 months of trading days
- BET reacts more to news ( $\hat{\alpha} = 0.192$ ) and forgets faster than the S&P 500
  - ▶ BET-TR (2014–2026 only): half-life 8 days; a shorter, calmer sample gives a lower persistence
- Bitcoin: the heaviest tails ( $\hat{\nu} = 3.18$ , fourth moment infinite) and a unit root in variance
- EUR/RON: a managed exchange rate; long quiet periods, then jumps
  - ▶ 6 May 2025: +1.20% after days of  $\pm 0.004\%$ : standardised residual of 154

## Persistence: How Fast Do Shocks Fade?



□ Share of a variance shock still present after  $h$  days,  $(\hat{\alpha} + \hat{\beta})^h$ ; dashed line: one half

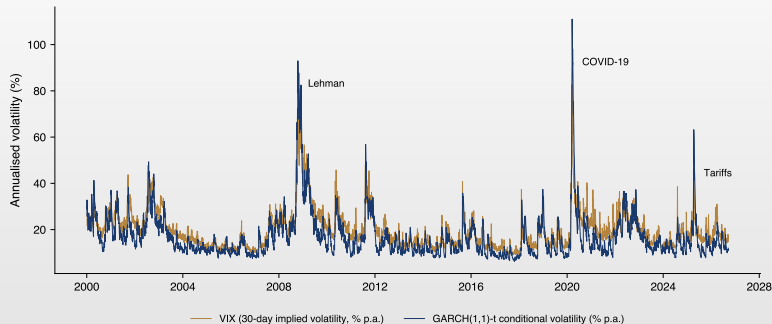
## Interpretation: Persistence

- Half-lives: S&P 500 131 days, gold 106, BET 74, BET-TR 8
- Bitcoin and EUR/RON lie on the flat line: an estimated  $\alpha + \beta$  of 1 means shocks never fade in the model
- Very high persistence can be spurious: level shifts in volatility mimic a unit root [Lamoureux & Lastrapes \(1990\)](#)
  - ▶ Regime-switching alternatives: [Hamilton & Susmel \(1994\)](#)
- Practical rule: report half-lives with a confidence interval (parametric bootstrap, Seminar 5)

## Delta Method for Long-Run Volatility and Half-Life

- For a smooth  $g(\theta)$ :  $\widehat{\text{s.e.}} g(\hat{\theta}) \approx \sqrt{\nabla g' \hat{V} \nabla g}$ , with  $\hat{V}$  the robust covariance of  $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$
- S&P 500, GARCH(1,1)-t: long-run volatility 27.4% p.a., s.e. 10.8 points: the gradient is scaled by  $1/(1 - \alpha - \beta)^2$
- Half-life  $h = \ln 0.5 / \ln p$ ,  $p = \alpha + \beta$ :  $\partial h / \partial p = -\ln 0.5 / [p (\ln p)^2] = 24\,789$  at  $\hat{p} = 0.9947$ 
  - ▶  $\widehat{\text{s.e.}}(\hat{p}) = 0.0047$ , so  $\widehat{\text{s.e.}}(\hat{h}) \approx 116$  days for  $\hat{h} = 131$ : a symmetric interval would include negative half-lives
- The linearisation fails because  $h(p)$  explodes as  $p \rightarrow 1$ 
  - ▶ Transform an interval for  $p$  instead ( $h$  is monotone):  $[0.9855; 1.0039]$  gives  $h \in [48; \infty)$
  - ▶ Or use a parametric bootstrap and a test simulated under  $H_0$  (Seminar 5, B7)

## S&P 500: GARCH Volatility and the VIX

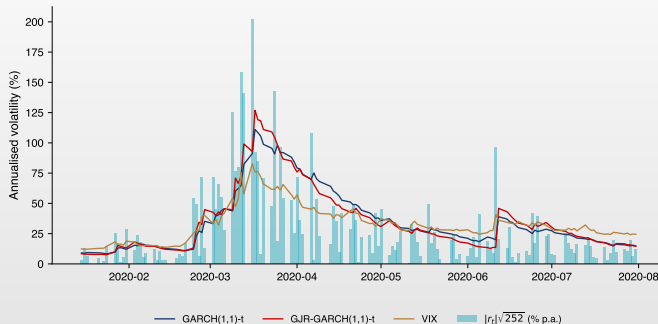


■ Annualised GARCH(1,1)-t conditional volatility  $\hat{\sigma}_t \sqrt{252}$  and the VIX, the 30-day implied volatility, 2000–2026

## Interpretation: Statistical versus Implied Volatility

- Correlation of the two series: 0.89; GARCH uses only past returns, the VIX uses option prices
- Average VIX 19.8% against average GARCH volatility 17.0%; VIX is higher on 84% of days
  - ▶ The gap is the **variance risk premium**: option buyers pay for insurance against volatility
- GARCH range: from 6.2% (October 2017) to 111% (March 2020)
- GARCH reacts one day late by construction: it needs the shock before it moves

## Zoom: The COVID-19 Crash, 2020



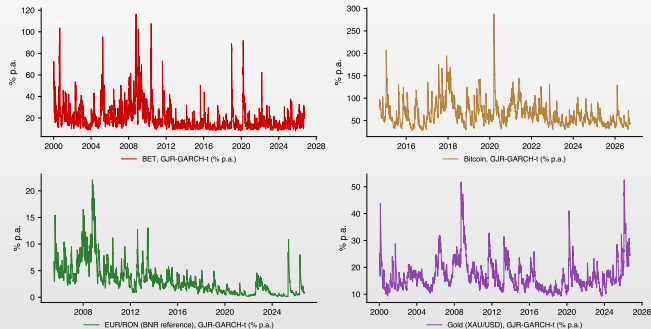
□ January–July 2020: absolute returns (annualised), GARCH-t and GJR-GARCH-t volatility, VIX

 MFM\_ch5\_sp500\_vix

## Interpretation: COVID-19

- ▣ 16 March 2020:  $r_t = -12.8\%$ , the largest daily fall of the sample; VIX closed at 82.7%
- ▣ Peak model volatility on 17 March: GARCH-t 111%, GJR-GARCH-t 127%
  - ▶ GJR goes higher because the shocks were negative (Section 5)
- ▣ 24 February 2020 ( $r_t = -3.41\%$ ): the VIX jumped the same day, from 17.1% to 25.0%
  - ▶ GARCH-t moved only the next day, from 12.2% to 22.5%: markets price news at once, GARCH only reacts
- ▣ After the peak, volatility decays at the rate  $\alpha + \beta$  per day, as the model predicts

## BET, Bitcoin, EUR/RON and Gold



- GJR-GARCH-t conditional volatility, annualised with each series' frequency (Bitcoin: 365 days)

## Interpretation: Four Markets

- ▣ BET: the 2008 crisis dominates; short, sharp spikes afterwards; current level 26.4%
- ▣ Bitcoin: volatility of 50–100% is normal; March 2020 is the highest peak
- ▣ EUR/RON: high volatility until 2012, then a long managed calm, with bursts in 2022 and 2025
  - ▶ Volatility close to zero on many days: a daily GARCH is a poor description of a managed rate
- ▣ Gold: peaks of 52% (2008), 31% (2013), 41% (2020) and a new high of 53% in 2026

## Recap: Results by Market

### Key points

- ▣ One-step and multi-step forecasts follow from the recursion; the 10-day variance is a sum of daily variances
- ▣ Persistence is high everywhere; half-lives range from about a week (BET-TR) to several months
- ▣ Estimated unit roots (Bitcoin, EUR/RON) make long-run forecasts meaningless
- ▣ GARCH tracks the VIX closely but lags it; the VIX is higher on average (variance risk premium)

## Bad News Moves Volatility More

- ▣ Stylised fact 9 (Chapter 1): negative returns raise future volatility more than positive returns of the same size
- ▣ **Leverage effect:** a falling price raises the debt-to-equity ratio, so equity becomes riskier
  - ▶ **Volatility feedback:** higher expected volatility raises required returns, so prices fall now
- ▣ GARCH(1,1) cannot see the sign:  $\varepsilon_{t-1}^2$  is the same for  $-3$  and  $+3$
- ▣ Two fixes: GJR adds a term for negative shocks, EGARCH models the log-variance

## GJR-GARCH (Glosten, Jagannathan and Runkle, 1993)

- $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ , with  $I_{t-1} = 1$  if  $\varepsilon_{t-1} < 0$  Glosten, Jagannathan & Runkle (1993)
  - ▶ Good news: slope  $\alpha$ ; bad news: slope  $\alpha + \gamma$ ;  $\gamma > 0$  means asymmetry
- Persistence with symmetric  $z_t$ :  $\alpha + \gamma/2 + \beta$  (half the shocks are negative)
  - ▶ Long-run variance  $\omega/(1 - \alpha - \gamma/2 - \beta)$
- Test of symmetry:  $H_0: \gamma = 0$  by a t-test or by LR against GARCH(1,1)
- S&P 500:  $\hat{\alpha} = 0.000$ ,  $\hat{\gamma} = 0.205$  ( $t = 10.0$ ),  $\hat{\beta} = 0.881$

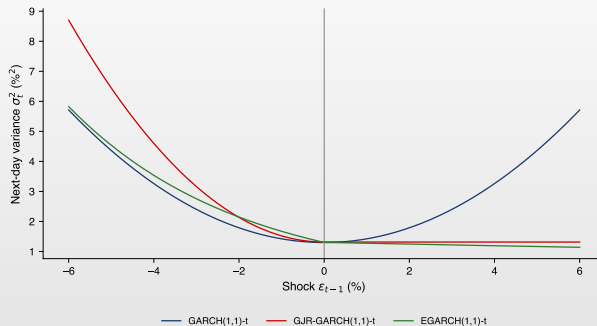
## EGARCH (Nelson, 1991)

- $\ln \sigma_t^2 = \omega + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$  Nelson (1991)
  - ▶ Size effect  $\alpha$ , sign effect  $\gamma$ ;  $\gamma < 0$ : negative shocks raise volatility more
- The log form keeps  $\sigma_t^2 > 0$  without any sign restriction on the parameters
- Persistence:  $\beta$  (stationary if  $|\beta| < 1$ )
- S&P 500:  $\hat{\alpha} = 0.138$ ,  $\hat{\gamma} = -0.165$  ( $t = -15.2$ ),  $\hat{\beta} = 0.979$
- In arch,  $\mathbb{E}|z|$  is set to  $\sqrt{2/\pi}$ , its value under the Normal distribution

## The News Impact Curve

- ▣ **News impact curve** Engle & Ng (1993):  $\sigma_t^2$  as a function of  $\varepsilon_{t-1}$ , with  $\sigma_{t-1}^2$  fixed at the sample variance
  - ▶ GARCH: a parabola symmetric around 0
  - ▶ GJR: two half-parabolas with different slopes
  - ▶ EGARCH: exponential in  $|\varepsilon|$ , tilted by  $\gamma$
- ▣ It isolates how each model turns one day of news into tomorrow's variance
- ▣ Engle and Ng also propose the sign-bias tests used in the diagnostics (Section 6)

## News Impact Curves for the S&P 500



- Next-day variance after a shock  $\varepsilon_{t-1}$  (in %), for the three fitted models with Student-t innovations

## Interpretation: News Impact

- ▣ Starting variance 1.47 (sample variance, %<sup>2</sup>)
- ▣ A shock of  $-3\%$ : GJR gives 3.16, EGARCH 2.76; a shock of  $+3\%$ : GJR 1.31, EGARCH 1.22
  - ▶ Bad news raises the variance 2.4 times more than good news in GJR, 2.3 times in EGARCH
- ▣ Symmetric GARCH gives 2.40 for both signs: too much after good news, too little after bad news
- ▣ For the S&P 500,  $\hat{\alpha} = 0$  in GJR: positive shocks do not raise volatility at all

## Asymmetry in Six Markets

| Series  | $\hat{\gamma}_{GJR}$ | t    | LR vs GARCH | p-value | $\hat{\gamma}_{EGARCH}$ | t     |
|---------|----------------------|------|-------------|---------|-------------------------|-------|
| S&P 500 | 0.205                | 10.0 | 235.7       | < 0.001 | -0.165                  | -15.2 |
| BET     | 0.043                | 2.1  | 4.9         | 0.026   | -0.028                  | -2.6  |
| BET-TR  | 0.111                | 2.9  | 10.3        | 0.001   | -0.063                  | -3.5  |
| Bitcoin | -0.013               | -0.7 | 0.7         | 0.416   | 0.015                   | 1.2   |
| EUR/RON | 0.039                | 3.0  | 9.3         | 0.002   | -0.040                  | -2.5  |
| Gold    | -0.043               | -4.7 | 27.6        | < 0.001 | 0.038                   | 5.7   |

- All models with Student-t innovations; t-statistics with robust standard errors;  $LR \sim \chi^2_1$  (critical value 3.84)
- Asymmetry: GJR  $\hat{\gamma} > 0$  and EGARCH  $\hat{\gamma} < 0$  for the same direction (bad news matters more)

## Interpretation: Where Is the Leverage Effect?

- S&P 500: very strong ( $LR = 235.7$ ); the classic leverage effect of a developed equity index
  - ▶ BET and BET-TR: present but weak ( $LR = 4.9$  and  $10.3$ )
- Bitcoin: no asymmetry ( $LR = 0.7$ ); crypto rallies are as volatile as crashes
- Gold: **inverted** asymmetry ( $\hat{\gamma}_{GJR} = -0.043$ ): price rises raise volatility more
  - ▶ Consistent with gold as a safe haven: it rallies in turbulent times
- EUR/RON: GJR  $\hat{\gamma} > 0$ , EGARCH  $\hat{\gamma} < 0$ : falls of the rate (leu appreciation) raise volatility slightly more

## Recap: Asymmetry

### Key points

- ▣ Symmetric GARCH ignores the sign of news; equity data reject this strongly
- ▣ GJR adds a slope  $\gamma$  for negative shocks; EGARCH models  $\ln \sigma_t^2$  with a sign term
- ▣ The news impact curve shows the asymmetry directly
- ▣ Asymmetry is an equity phenomenon: absent for Bitcoin, reversed for gold

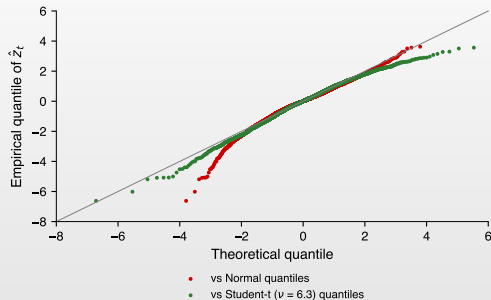
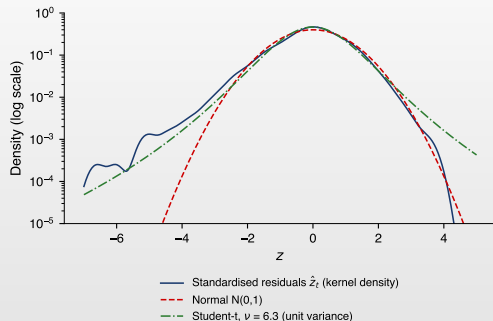
## Standardised Residuals: What Should Hold

- ▣ **Standardised residual:**  $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ ; if the model is right,  $\hat{z}_t$  is close to i.i.d. with variance 1
- ▣ Checks
  - ▶ No autocorrelation in  $\hat{z}_t$ : Ljung–Box  $Q(10)$  on  $\hat{z}_t$  (mean equation)
  - ▶ No clustering left:  $Q(10)$  on  $\hat{z}_t^2$  and the LM(5) test (variance equation)
  - ▶ No asymmetry left: the sign-bias tests [Engle & Ng \(1993\)](#)
  - ▶ Distribution: compare  $\hat{z}_t$  with the assumed density (QQ plot)

## The Sign-Bias Tests (Engle and Ng, 1993)

- Regress  $\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{z}_{t-1} + b_3 S_{t-1}^+ \hat{z}_{t-1} + u_t$ 
  - ▶  $S_{t-1}^- = 1$  if  $\hat{z}_{t-1} < 0$ , else 0;  $S_{t-1}^+ = 1 - S_{t-1}^-$
- **Sign bias** ( $b_1$ ): do negative shocks alone raise the next squared residual?
- **Negative / positive size bias** ( $b_2, b_3$ ): does the size of negative / positive shocks matter beyond the model?
- Joint test:  $nR^2 \sim \chi_3^2$  (5% critical value 7.81)

## S&P 500: Are the Innovations Normal?



- Standardised residuals of GARCH(1,1) with Normal innovations: log-density and QQ plot against Normal and Student-t

## Interpretation: The Tails After GARCH

- Excess kurtosis drops from 10.7 (returns) to 1.73 (standardised residuals): GARCH explains most of it
- But not all: Jarque–Bera 1153; skewness -0.53
  - ▶ 15 residuals below  $-4$ ; the Normal distribution expects 0.21, a Student-t with  $\nu = 6.3$  expects 8.5
  - ▶ Largest:  $\hat{z} = -6.6$  on 27 February 2007, a large fall in a calm period
- The left tail is longer than the right: skewed-t with  $\hat{\lambda} = -0.150$  (GJR,  $\hat{\eta} = 7.7$ )
- Hence BIC prefers skewed-t innovations for the S&P 500

## S&P 500: Diagnostics of Five Models

| Model        | $Q(10)$ on $\hat{z}$ | $Q(10)$ on $\hat{z}^2$ | LM(5) | Sign bias $t$ | Joint $\chi^2_3$ |
|--------------|----------------------|------------------------|-------|---------------|------------------|
| GARCH-N      | 21.2                 | 16.4                   | 8.0   | 2.59          | 44.5             |
| GARCH-t      | 21.9                 | 13.6                   | 5.8   | 2.47          | 39.3             |
| GJR-t        | 22.3                 | 10.2                   | 4.5   | 3.96          | 23.5             |
| EGARCH-t     | 27.6                 | 10.5                   | 8.2   | 2.95          | 8.8              |
| GJR-skewed-t | 19.7                 | 10.7                   | 5.0   | 4.22          | 27.1             |

- 5% critical values:  $\chi^2_{10}$ : 18.3;  $\chi^2_5$ : 11.1;  $\chi^2_3$ : 7.81;  $|t| > 1.96$
- Before modelling,  $Q(10)$  on  $r_t^2$  was 6133

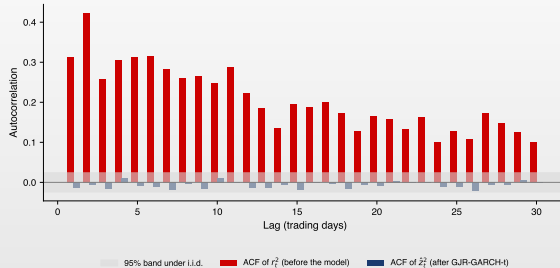
## Interpretation: What the Diagnostics Say

- Clustering is gone for all models:  $Q(10)$  on  $\hat{z}^2$  falls from 6133 to 10–16,  $LM(5)$  is not significant
- Symmetric GARCH fails the sign-bias test (joint 39.3): asymmetry left over
  - ▶ GJR still fails (23.5); EGARCH comes closest (8.8,  $p = 0.032$ )
- $Q(10)$  on  $\hat{z}$  around 20–28: some autocorrelation in the mean, which a constant mean ignores
- With  $n = 6\,714$ , small effects are significant: judge economic size, not only p-values

## Residual Diagnostics Have Estimated Parameters

- $\hat{z}_t$  depends on  $\hat{\theta}$ : the Ljung–Box statistic on  $\hat{z}_t^2$  is not exactly  $\chi_m^2$  Li & Mak (1994)
  - ▶ Squared-residual autocorrelations  $\hat{r}_k$ :  $\sqrt{n} \hat{r} \rightarrow N(0, C)$ ,  $C = I - H' J^{-1} H / (\kappa_z - 1)$
  - ▶  $H_k = \mathbb{E}[g_t(z_{t-k}^2 - 1)]$ : how the fitted variance absorbs lag- $k$  dependence; the ARMA rule  $\chi_{m-p-q}^2$  does not apply
- S&P 500, Gaussian QMLE of GARCH(1,1):  $C_{1,1} = 0.78$ ,  $C_{5,5} = 0.93$ ,  $C_{10,10} = 0.97$ 
  - ▶ The correction is largest at small lags, where  $\hat{\alpha}$  has already fitted the dependence
- $Q(10)$  on  $\hat{z}_t^2$ : Ljung–Box 16.4 ( $p = 0.089$ ); corrected  $n\hat{r}'C^{-1}\hat{r} = 19.1$  ( $p = 0.038$ ): the 5% verdict flips
- The naive test is conservative: it under-rejects a misspecified variance equation; the joint sign-bias test carries the same caveat

## Diagnostics: Squared Returns versus Squared Residuals



- S&P 500: autocorrelations of  $r_t^2$  and of  $\hat{z}_t^2$  from GJR-GARCH-t, lags 1–30
- After the model, the autocorrelations sit inside the i.i.d. band: the variance dynamics are captured

## Diagnostics in Other Markets

- BET:  $Q(10)$  on  $\hat{z} = 116.6$ : strong autocorrelation in the mean (thin trading, Chapter 2)
  - ▶ Remedy: an AR(1) mean equation,  $r_t = \mu + \phi r_{t-1} + \varepsilon_t$
- Bitcoin: variance equation passes ( $Q(10)$  on  $\hat{z}^2 = 4.7$ ), no sign bias (joint 2.3)
- EUR/RON: residual excess kurtosis 3190, driven by the single residual of 154 (May 2025)
  - ▶ One observation can dominate the diagnostics: always look at the largest residuals
- Gold: EGARCH leaves clustering ( $Q(10)$  on  $\hat{z}^2 = 48.8$ ) although its BIC is among the best

## Recap: Innovations and Diagnostics

### Key points

- ▣ Standardised residuals must look i.i.d.: test their level, their squares and their signs
- ▣ GARCH removes the clustering and most of the kurtosis, not the skewness
- ▣ Heavy-tailed, skewed innovations and asymmetric variance are both needed for the S&P 500
- ▣ Diagnostics can disagree with information criteria; one extreme residual can drive them

## The Proxy Problem

- Volatility is never observed: we compare forecasts with a noisy **proxy**, e.g.  $r_t^2$
- $r_t^2 = \sigma_t^2 z_t^2$  is unbiased ( $\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$  for  $\mu = 0$ ) but very noisy
  - ▶ Its  $R^2$  against the true variance is small even for a perfect forecast [Andersen & Bollerslev \(1998\)](#)
- [Patton \(2011\)](#): only some loss functions rank forecasts correctly with a noisy proxy
  - ▶ Robust: MSE  $(r_t^2 - h_t)^2$  and **QLIKE**  $r_t^2/h_t + \ln h_t$ , with  $h_t$  the variance forecast
  - ▶ Not robust: losses on volatility  $|r_t| - \sqrt{h_t}$  or on logs
- QLIKE is (up to constants) minus the Normal log-likelihood; it penalises under-prediction more

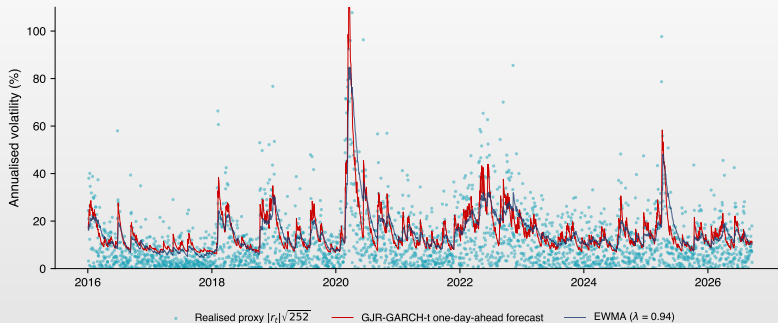
## Mincer–Zarnowitz and Diebold–Mariano

- **Mincer–Zarnowitz regression** Mincer & Zarnowitz (1969):  $r_t^2 = a + b h_t + u_t$ 
  - ▶ Unbiased forecast:  $a = 0$ ,  $b = 1$  (joint Wald test with HAC errors);  $R^2$  measures information content
  - ▶  $b < 1$ : forecasts too dispersed (too high when high, too low when low)
- **Diebold–Mariano test** Diebold & Mariano (1995): loss differential  $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$ 
  - ▶  $DM = \bar{d} / \widehat{\text{s.e.}}(\bar{d})$  with a HAC standard error;  $\approx N(0, 1)$  under equal accuracy
  - ▶  $DM < -1.96$ : model A significantly better

## Out-of-Sample Design

- ▣ S&P 500; forecast period 2016–2026: 2 692 one-day-ahead forecasts
- ▣ Parameters re-estimated at the start of each year on all data since 2000 (expanding window)
  - ▶ Within the year, the variance is updated daily with fixed parameters: no look-ahead
- ▣ Models: GARCH with Normal and Student-t innovations, GJR-t, EGARCH-t
- ▣ Benchmarks: EWMA ( $\lambda = 0.94$ ) and the 20-day historical variance (Hist-20)
- ▣ Also 22-day (one-month) cumulative variance forecasts on 122 non-overlapping windows

## Forecasts versus Realised Moves, 2016–2026



One-day-ahead volatility forecasts (annualised) of GJR-GARCH-t and EWMA; teal dots:  $|r_t|\sqrt{252}$

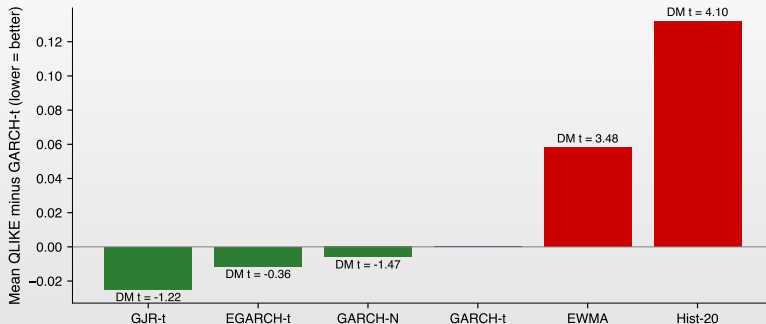
## S&amp;P 500: Forecast Evaluation, 2016–2026

| Model    | QLIKE | MSE  | DM t vs GARCH-t | MZ b | MZ $R^2$ | Wald p |
|----------|-------|------|-----------------|------|----------|--------|
| GARCH-N  | 0.751 | 22.3 | -1.47           | 0.97 | 0.250    | 1.00   |
| GARCH-t  | 0.757 | 22.8 | —               | 0.88 | 0.238    | 0.80   |
| GJR-t    | 0.732 | 21.8 | -1.22           | 0.84 | 0.276    | 0.68   |
| EGARCH-t | 0.745 | 21.3 | -0.36           | 1.24 | 0.293    | 0.71   |
| EWMA     | 0.815 | 24.7 | 3.48            | 0.89 | 0.172    | 0.69   |
| Hist-20  | 0.889 | 26.1 | 4.10            | 0.68 | 0.152    | 0.03   |

- Mean losses on 2 692 days; DM on QLIKE, HAC with 5 lags; negative t: better than GARCH-t
- MZ:  $r_t^2 = a + b h_t$ ; Wald p: test of  $a = 0$ ,  $b = 1$

 MFM\_ch5\_forecast\_evaluation

## QLIKE Relative to GARCH-t



- Mean QLIKE of each model minus that of GARCH-t, with the Diebold–Mariano t-statistic

## Interpretation: Does Anything Beat GARCH(1,1)?

- Best QLIKE: GJR-t (0.732); but the GARCH-family differences are not significant (DM t from -1.22 to -0.36)
- EWMA and Hist-20 are significantly worse (DM t = 3.48 and 4.10)
  - ▶ Mean reversion and estimated parameters pay off
- MZ: GARCH-type forecasts are unbiased ( $b$  close to 1, Wald  $p > 0.6$ ); Hist-20 is not ( $p = 0.03$ )
- $R^2$  of only 0.15–0.29: the noise of  $r_t^2$ , not a bad model
- Same message as [Hansen & Lunde \(2005\)](#): GARCH(1,1) is hard to beat; asymmetry helps a little for equities

## Comparing Forecasts from Estimated, Nested Models

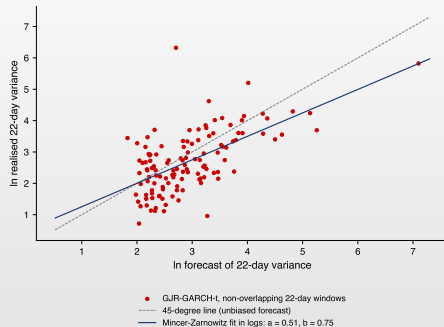
- DM treats forecasts as given; ours come from estimated parameters, and GARCH-t is nested in GJR-t
  - ▶ Expanding window, nested models: under  $H_0$  the loss differential degenerates, so DM is not  $N(0, 1)$  [West \(1996\)](#), [Clark & McCracken \(2001\)](#)
- [Giacomini & White \(2006\)](#): test the forecasting **method** (model + rolling window of fixed length  $W$ )
  - ▶ Valid for nested models and estimated parameters, because estimation noise does not vanish
  - ▶ Conditional test  $\mathbb{E}[d_{t+1} | \mathcal{F}_t] = 0$  with instruments  $h_t = (1, d_t)$ :  $GW = n \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \rightarrow \chi^2_2$ ,  $Z_{t+1} = h_t d_{t+1}$
- Six models and five pairwise DM tests: a multiple-comparison problem
  - ▶ **Model Confidence Set** [Hansen, Lunde & Nason \(2011\)](#): drop the worst model until equal accuracy is not rejected; the set contains the best model with probability  $\geq 1 - \alpha$

## S&P 500: Giacomini–White Tests and the Model Confidence Set

| Model    | QLIKE (W) | $t$ uncond. | GW    | $p$     | MCS $p$ |
|----------|-----------|-------------|-------|---------|---------|
| GARCH-N  | 0.751     | 1.87        | 5.85  | 0.054   | 0.586   |
| GARCH-t  | 0.743     | —           | —     | —       | 0.353   |
| GJR-t    | 0.738     | -0.18       | 2.49  | 0.288   | 1.000   |
| EGARCH-t | 0.746     | 0.07        | 3.21  | 0.201   | 0.586   |
| EWMA     | 0.815     | 3.20        | 11.40 | 0.003   | 0.008   |
| Hist-20  | 0.889     | 3.87        | 16.41 | < 0.001 | 0.001   |

- Rolling window  $W = 1\,000$  days, re-estimated each January, 2 692 forecasts; tests against GARCH-t, HAC with 5 lags
- MCS on the expanding-window forecasts of Section 7: QLIKE, statistic  $T_R$ , stationary bootstrap (mean block 10, 10 000 draws)
- 90% MCS: GARCH-N, GARCH-t, GJR-t, EGARCH-t; EWMA and Hist-20 excluded: the same verdict as DM, now with a controlled error rate

## One-Month Forecasts: Mincer–Zarnowitz in Logs



- GJR-GARCH-t forecast of the 22-day variance against the realised sum of 22 squared returns, 122 non-overlapping windows

## Interpretation: Monthly Horizon

- ▣ Aggregating 22 days reduces the proxy noise:  $R^2$  in logs 0.38 (GJR-t), against 0.28 for one day in levels
- ▣ Slope  $b = 0.75$  (s.e. 0.07): monthly forecasts are too dispersed
  - ▶ High forecasts are followed by less variance than predicted: consistent with faster mean reversion than  $\alpha + \beta$  implies
- ▣ In levels the slope is only 0.27: two COVID-19 windows (February and March 2020) dominate the regression
- ▣ Evaluate on logs or with QLIKE when a few crisis months carry huge variances

## Recap: Forecasting

### Key points

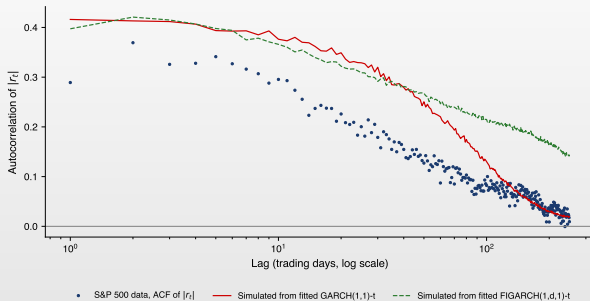
- ▣ Evaluate forecasts out of sample, with parameters estimated only on past data
- ▣ With a noisy proxy use robust losses (QLIKE, MSE) and the Diebold–Mariano test
- ▣ Nested models with estimated parameters: Giacomini–White with a rolling window; many models: Model Confidence Set
- ▣ Mincer–Zarnowitz checks unbiasedness; low  $R^2$  at one day is expected
- ▣ GARCH beats EWMA and historical variance; members of the GARCH family are hard to separate

## GARCH-in-Mean: Is Risk Rewarded?

- **GARCH-M** Engle, Lilien & Robins (1987):  $r_t = c + \kappa \sigma_t + \varepsilon_t$  (or  $\kappa \sigma_t^2$ )
  - ▶  $\kappa > 0$ : investors demand a higher expected return when risk is higher (risk-return trade-off)
- S&P 500, GARCH(1,1)-t in mean:  $\hat{\kappa} = 0.096$  (t = 3.02) with  $\sigma_t$ ; 0.028 (t = 2.35) with  $\sigma_t^2$ 
  - ▶ Positive and significant in this sample, but the literature is mixed: GJR (1993) found a negative relation
- Caution: a daily mean effect is small and fragile; it is sensitive to the sample and to the asymmetry in variance

 MFM\_ch5\_long\_memory

## Long Memory: Where GARCH Falls Short



- S&P 500: autocorrelation of  $|r_t|$  (lags 1–250, log scale) in the data and in long paths simulated from the fitted GARCH(1,1)-t and FIGARCH(1,d,1)-t

## Interpretation: Long Memory

- Data: 0.289 at lag 1, 0.082 at lag 100, 0.009 at lag 250: a slow, roughly log-linear decline
  - ▶ GARCH: too high at short lags (0.416), then an exponential collapse: its ACF has a characteristic time
- **FIGARCH** Baillie, Bollerslev & Mikkelsen (1996): fractional integration  $(1 - L)^d$  gives hyperbolic decay
  - ▶  $\hat{d} = 0.60$  (s.e. 0.06); BIC 34.2 lower than GARCH, but the simulated memory is now too long (0.144 at lag 250)
- No single-scale model fits all horizons: Chapter 9 uses realised volatility and the HAR model

## Other Limits of GARCH

- ▣ **Structural breaks:** shifts in the volatility level inflate  $\alpha + \beta$  towards 1 [Lamoureux & Lastrapes \(1990\)](#)
  - ▶ Our BET-TR (2014–2026) persistence 0.9188 vs BET (2000–2026) 0.9907: the sample matters
- ▣ **Reaction lag:** GARCH responds after the shock; implied volatility, e.g. the VIX, moves before
- ▣ **Daily data only:** intraday information is lost; realised measures use it (Chapter 9)
- ▣ **One asset at a time:** portfolios need covariances; multivariate GARCH and DCC in Chapter 6
- ▣ **Managed prices** (EUR/RON): a daily GARCH describes interventions poorly

## Beyond GARCH(1,1): Score-Driven, Realized and Component Models

- **Score-driven (GAS)** Creal, Koopman & Lucas (2013):  $\sigma_{t+1}^2 = \omega + A s_t + B \sigma_t^2$ ,  $s_t$  the scaled score of the Student-t density
  - ▶ The score is bounded in  $z_t$ : an outlier such as the EUR/RON residual of 154 moves the variance far less than in GARCH
- **Realized GARCH** Hansen, Huang & Shek (2012): adds a measurement equation linking  $\sigma_t^2$  to a realised variance
  - ▶ Uses intraday information and reacts faster; the bridge to Chapter 9
- **GARCH-MIDAS** Engle, Ghysels & Sohn (2013):  $\sigma_t^2 = \tau_t g_t$ , a slow long-run component  $\tau_t$  driven by macro data (mixed frequencies)
  - ▶ A moving long-run level absorbs the level shifts that push  $\hat{\alpha} + \hat{\beta}$  towards 1
- Each extension is judged like GARCH: out of sample, robust loss, Model Confidence Set

## Beyond Variance: Entropy and Tail Risk

- [Pele, Lazar & Dufour \(2017\)](#): entropy of the intraday return distribution and market risk (EUR/JPY)
  - ▶ Higher entropy goes with lower intraday Value at Risk and expected shortfall; entropy then forecasts daily Value at Risk
- [Pele & Mazurencu-Marinescu-Pele \(2019\)](#): entropy of intraday Bitcoin returns as a predictor of daily Value at Risk
  - ▶ In backtests, the entropy-based forecast beats the classical GARCH models
- [Pele, Lazar & Mazurencu-Marinescu-Pele \(2019\)](#): a nonparametric tail measure based on entropy, compared with standard expected shortfall models
- GARCH volatility feeds directly into Value at Risk and expected shortfall: Chapters 7–8

## An Open Question

- Do machine-learning or realised-measure forecasts beat a well-specified GARCH for daily equity variance, out of sample, once the number of models compared is accounted for?
- The question is live
  - ▶ Christensen, Siggaard & Veliyev (2023): machine-learning methods with little tuning are competitive with HAR-type models for the realised variance of Dow Jones stocks
  - ▶ Hansen, Lunde & Nason (2011): the Model Confidence Set keeps every model not significantly worse than the best, the right object when many models compete
  - ▶ This chapter: within the GARCH family, the DM statistic ranges from  $-1.22$  to  $-0.36$ : no significant winner
- Formal statement for models A and B, QLIKE loss  $L$ 
  - ▶  $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$ ;  $H_0: \mathbb{E}[d_t] = 0$ ;  $H_1: \mathbb{E}[d_t] < 0$ ; DM statistic with HAC errors Diebold & Mariano (1995)
  - ▶ Falsified if A leaves the Model Confidence Set, or if the sign of  $\bar{d}$  flips across pre-registered subperiods

## The Discovery Loop, Step by Step (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List out-of-sample comparisons of GARCH, HAR and machine-learning variance forecasts since 2015, with DOI, asset, loss function, test and horizon."*
  - ▶ Resolve every DOI on Crossref; check that the loss is robust to a noisy proxy [Patton \(2011\)](#)
- **Hypotheses:** *"For the claim that asymmetry improves one-day forecasts, list competing mechanisms (leverage, volatility feedback, a few crisis days) and what each predicts for  $d_t$  given the sign of  $r_{t-1}$  and by subperiod."*
- **Data and code:** *"Write Python with `arch` that re-estimates GARCH-t and GJR-t each January on an expanding window from data/market and stores one-day variance forecasts; no parameter may use data after the forecast origin."*
  - ▶ Replicate first: QLIKE 0.757 (GARCH-t) and 0.732 (GJR-t), DM  $t = -1.22$  (Section 7)

## The Discovery Loop, Step by Step (2/2)

- **Robustness:** *“Write a pre-registered plan: splits by the sign of  $r_{t-1}$  and by subperiod, MSE besides QLIKE, the 22-day horizon, the HAC lag, and a Bonferroni or Model Confidence Set correction.”*
  - ▶ Placebo: random splits of the same sizes; the gap in  $\bar{d}$  between down-day and up-day subsets must exceed the 95% band of random gaps
- **Critique:** *“Act as a referee: could the proxy  $r_t^2$ , one crisis year or the HAC lag create my result? Give the test that would expose each.”*
- **Report:** all splits and losses tried, adjusted  $p$ -values, code, seeds and package versions
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

## What the Researcher Must Check

- **Look-ahead:** parameters fitted on the full sample, or  $h_t$  aligned with  $r_t$  instead of  $r_{t-1} \Rightarrow$  assert the last estimation date precedes each origin; shift the forecast one day and compare
- **Non-robust loss:** losses on  $|r_t|$  or on logs can rank forecasts wrongly with a noisy proxy [Patton \(2011\)](#)  $\Rightarrow$  QLIKE and MSE only
- **Model mining:** 3 variance equations  $\times$  4 distributions  $\times$  windows  $\Rightarrow$  Model Confidence Set or Bonferroni over the pre-registered set
- **Influential periods:** a crisis year can dominate  $\bar{d} \Rightarrow$  report subperiods, and state in advance whether any year may be excluded
- **Scale and convergence:** returns in % vs decimals, failed optimisations  $\Rightarrow$  check the optimiser flags and one variance update step by step
- **Invented references or functions:** unresolvable DOIs, arch arguments that do not exist  $\Rightarrow$  Crossref and running the code; log in `AI_ERRORS.md`

## Mini-Case: Does Asymmetry Improve the Forecast?

- A typical assistant proposal: "GJR-t beats GARCH-t because of the leverage effect";  $d_t = \text{QLIKE}(\text{GJR-t}) - \text{QLIKE}(\text{GARCH-t})$
- Replicate: 2 692 days, 2016–2026:  $\bar{d} = -0.025$ , DM  $t = -1.22$  ( $p = 0.22$ ): not significant
- Testable implication of leverage: the gain sits after down days
  - ▶ After  $r_{t-1} < 0$  (1 223 days):  $\bar{d} = -0.015$ ,  $t = -0.51$ ; after  $r_{t-1} \geq 0$  (1 469):  $\bar{d} = -0.034$ ,  $t = -1.12$
  - ▶ Not supported: GJR has  $\hat{\alpha} = 0.000$ , so a rise does not raise its forecast; the gain is no smaller after up days
- Subperiods: 2016–2020  $t = -0.02$ ; 2021–2026  $t = -4.06$  ( $p < 0.001$ ); without 2020  $t = -4.26$ 
  - ▶ The 2021–2026 gain survives Bonferroni over the 5 splits ( $0.05/5 = 0.01$ ); dropping 2020 is legitimate only if pre-registered
- Interpretation: asymmetry helps, but not through the channel proposed, and not in 2016–2020

## Your Turn: A Team-Project Seed

- Question: do realised-variance (HAR) or machine-learning forecasts enter the Model Confidence Set together with GARCH, GJR and EGARCH for the S&P 500 and the BET, 2016–2026?
- Replicate first: the QLIKE table of Section 7 (GARCH-t 0.757, EWMA DM  $t = 3.48$ ) and the mini-case splits
- Deliverables
  - ▶ A one-page pre-registration: models, losses, splits, error rate and what would falsify the claim
  - ▶ Code on data/market (daily and intraday), the Model Confidence Set at 10%, a referee-style self-critique
- Log every prompt, tool and accepted output in `AI_USE.md`; every error caught (look-ahead, wrong scale, invented DOI) and the test that caught it in `AI_ERRORS.md`
- Code of the mini-case: [Quantlets/Ch\\_05/ai\\_discovery\\_case.py](#)

## Key Takeaways

### What to remember

- ▣ Returns are nearly unpredictable, their size is not: GARCH turns clustering into forecasts
- ▣ GARCH(1,1):  $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ ; persistence  $\alpha + \beta$  close to 1 in all our markets
- ▣ Equity volatility is asymmetric (GJR, EGARCH); Bitcoin is not; gold is reversed
- ▣ Innovations are heavy-tailed and skewed: Student-t or skewed-t, with robust standard errors
- ▣ Out of sample, GARCH models beat EWMA and historical variance; among themselves, differences are small

## A Practical Checklist

- ▣ 1. Test for ARCH effects:  $Q$  on  $r_t^2$  and the LM test
- ▣ 2. Fit GARCH, GJR and EGARCH with Normal, Student-t and skewed-t innovations; returns in %
- ▣ 3. Check convergence, bounds and robust standard errors (valid only if  $\mathbb{E}z_t^4 < \infty$ ); boundary-adjusted LR and BIC
- ▣ 4. Diagnose the standardised residuals (level, squares, signs, largest values)
- ▣ 5. Evaluate forecasts out of sample: QLIKE, Diebold–Mariano or Giacomini–White, Model Confidence Set, Mincer–Zarnowitz
- ▣ 6. Report the half-life with its uncertainty; distrust long-run numbers when  $\alpha + \beta \approx 1$

## Further Resources

### ▣ Books

- ▶ [Franke, Härdle & Hafner \(2019\)](#), chapters on ARIMA and GARCH models for financial time series
- ▶ [Francq & Zakoïan \(2019\)](#): the mathematical theory of GARCH models

### ▣ Quantinar courses

- ▶  [Statistics of Financial Markets](#)
- ▶  [Applied Time Series Analysis with Python](#)

### ▣ Code

- ▶ All charts and tables: [Quantlets/Ch\\_05](#); Python package [Sheppard, arch](#)

## Looking Ahead

- ▣ Chapter 6: dependence between markets, multivariate volatility and dynamic correlations
- ▣ Chapter 7: Value at Risk and expected shortfall, with GARCH volatility as an input
- ▣ Chapter 8: backtesting the risk forecasts
- ▣ Chapter 9: realised volatility and long memory
- ▣ Seminar 5: multi-step variance, delta method and QMLE efficiency derivations, boundary LR, Giacomini–White and MCS, IGARCH test by bootstrap, US spillovers to the BET

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